PSTAT220A Homework 2

Due 10/25/2026

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Problem 1 (35 pt)

The dataset `Salaries` in library `carData` concerns the salaries for Professors in 2008-2009. Please read the `Salaries` documentation for a full description of the data.

a) Numerical and Graphical summary of the data

```
data(Salaries)
head(Salaries)
```

```
       rank discipline yrs.since.phd yrs.service  sex salary
1      Prof          B            19          18 Male 139750
2      Prof          B            20          16 Male 173200
3  AsstProf          B             4           3 Male  79750
4      Prof          B            45          39 Male 115000
5      Prof          B            40          41 Male 141500
6 AssocProf          B             6           6 Male  97000
```

```
summary(Salaries)
```

```
       rank     discipline yrs.since.phd    yrs.service        sex
 AsstProf : 67   A:181      Min.   : 1.00   Min.   : 0.00   Female: 39
 AssocProf: 64   B:216      1st Qu.:12.00   1st Qu.: 7.00   Male  :358
 Prof     :266              Median :21.00   Median :16.00
                            Mean   :22.31   Mean   :17.61
                            3rd Qu.:32.00   3rd Qu.:27.00
                            Max.   :56.00   Max.   :60.00
    salary
 Min.   : 57800
 1st Qu.: 91000
```

```
 Median :107300
 Mean   :113706
 3rd Qu.:134185
 Max.   :231545
```

```
Salaries %>% group_by(rank) %>% summarise(n = n(), mean = mean(salary), sd = sd(salary))
```

```
# A tibble: 3 x 4
  rank          n    mean     sd
  <fct>     <int>   <dbl>  <dbl>
1 AsstProf     67  80776.  8174.
2 AssocProf    64  93876. 13832.
3 Prof        266 126772. 27719.
```

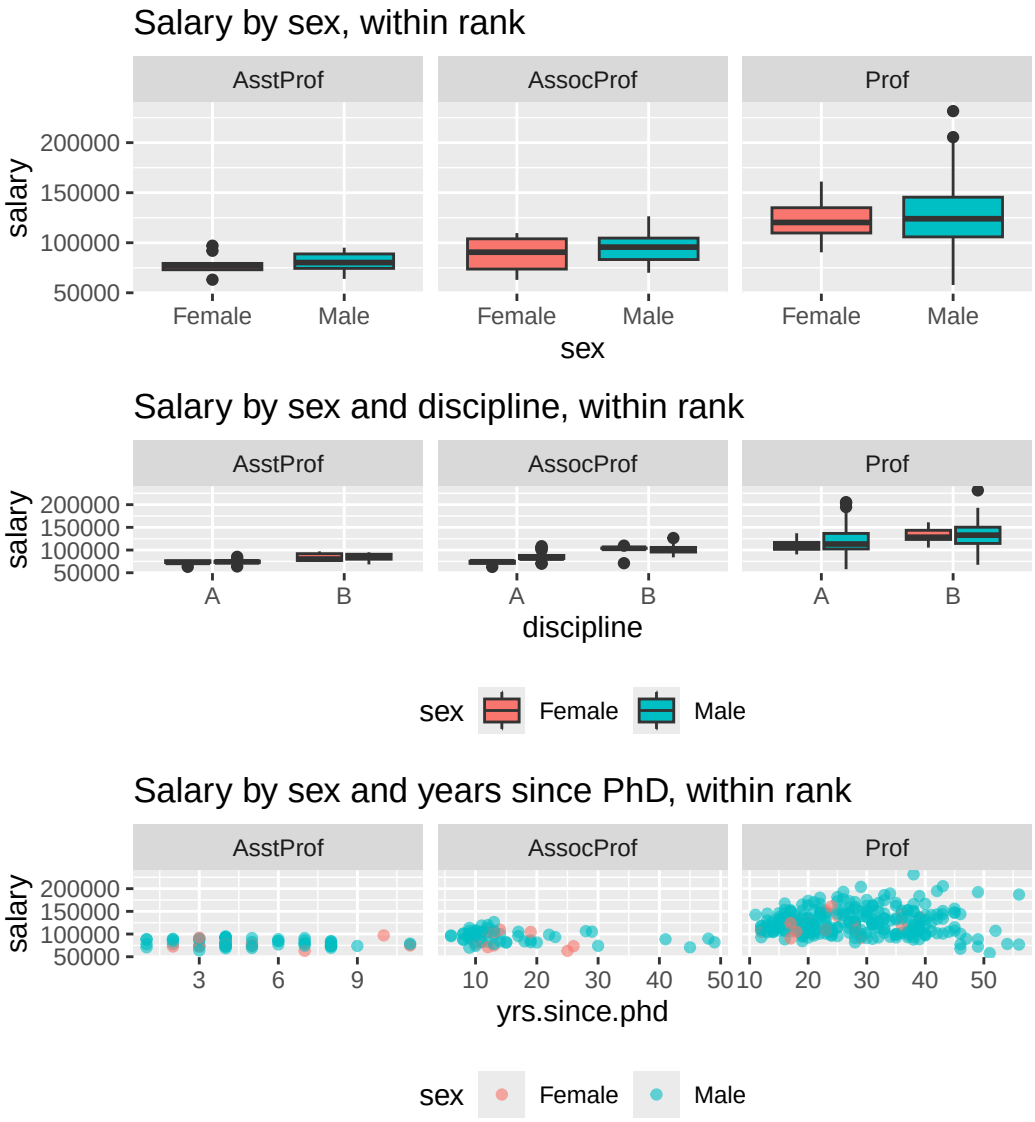
```
Salaries %>% group_by(sex) %>% summarise(n = n(), mean = mean(salary), sd = sd(salary))
```

```
# A tibble: 2 x 4
  sex        n    mean     sd
  <fct>  <int>   <dbl>  <dbl>
1 Female    39 101002. 25952.
2 Male     358 115090. 30437.
```

```
Salaries %>% group_by(discipline) %>% summarise(n = n(), mean = mean(salary), sd = sd(salary
```

```
# A tibble: 2 x 4
  discipline     n    mean     sd
  <fct>      <int>   <dbl>  <dbl>
1 A            181 108548. 30538.
2 B            216 118029. 29459.
```

```
p1 <- ggplot(Salaries, aes(sex, salary, fill = sex)) + geom_boxplot() +
  facet_wrap(~ rank) + labs(title = "Salary by sex, within rank") +
  theme(legend.position = "none")
p2 <- ggplot(Salaries, aes(discipline, salary, fill = sex)) + geom_boxplot() +
  facet_wrap(~ rank) + labs(title = "Salary by sex and discipline, within rank") +
  theme(legend.position = "bottom")
p3 <- ggplot(Salaries, aes(yrs.since.phd, salary, colour = sex)) + geom_point(alpha = .6) +
  facet_wrap(~ rank, scales = "free_x") +
  labs(title = "Salary by sex and years since PhD, within rank") +
  theme(legend.position = "bottom")
gridExtra::grid.arrange(p1, p2, p3, ncol = 1)
```



Comments. * Sampling bias: The sample is unbalanced. We have more data from men (358) than women (39), and more data from professors. * Sex bias: The median salary level is very similar, no matter the rank, whether asstprof, assocprof, or prof, but the variance is different. Specifically, within the ranks of assocprof and prof, the variance for females is a little narrower than for males, but for assocprof, it is broader than male. * Rank bias: We can find there is a very clear income rank pattern: prof > assoc prof > asst prof. * Discipline bias: We can not only find that the salary of discipline B is always higher than that of discipline A, but the variance also increases as the rank increases. * Years since PhD bias: We can find two interesting findings. Firstly, most asstprofs are within 0–10 years after their PhD, most assocprofs are concentrated within 10–20 years after their PhD, and most profs are within 20–40 years after their PhD. Secondly, it is straightforward that higher ranks usually align with higher income variance.

b) Linear regression

```r
fit <- lm(salary ~ rank + discipline + yrs.since.phd + yrs.service + sex, data = Salaries)
summary(fit)
```

```
Call:
lm(formula = salary ~ rank + discipline + yrs.since.phd + yrs.service +
    sex, data = Salaries)

Residuals:
   Min     1Q Median     3Q    Max
-65248 -13211  -1775  10384  99592

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    65955.2     4588.6  14.374  < 2e-16 ***
rankAssocProf  12907.6     4145.3   3.114  0.00198 **
rankProf       45066.0     4237.5  10.635  < 2e-16 ***
disciplineB    14417.6     2342.9   6.154 1.88e-09 ***
yrs.since.phd    535.1      241.0   2.220  0.02698 *
yrs.service     -489.5      211.9  -2.310  0.02143 *
sexMale         4783.5     3858.7   1.240  0.21584
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 22540 on 390 degrees of freedom
Multiple R-squared:  0.4547,    Adjusted R-squared:  0.4463
F-statistic:  54.2 on 6 and 390 DF,  p-value: < 2.2e-16
```

At the 5% significance level, we find that rank, discipline, years since PhD, and years of service are significantly associated with salary. Professors with higher ranks and those in discipline B tend to earn more. Years since PhD is positively associated with salary, while years of service is negatively associated with salary one potential explaination is that there maybe colinerity correlation between years of service and years since PhD. Moreover, we do not find a significant salary difference between males and females after controlling for other variables.

c) Matrix LS solution

```r
X <- model.matrix(fit)
y <- Salaries$salary

# B_hat = (X'X)^{-1} X'y
beta_hat <- solve(t(X) %*% X) %*% t(X) %*% y
cbind(matrix = drop(beta_hat), lm = coef(fit))
```

```
                 matrix         lm
(Intercept)   65955.2324 65955.2324
rankAssocProf 12907.5879 12907.5879
rankProf      45065.9987 45065.9987
disciplineB   14417.6256 14417.6256
yrs.since.phd   535.0583   535.0583
yrs.service    -489.5157  -489.5157
sexMale         4783.4928  4783.4928
```

```r
all.equal(drop(beta_hat), coef(fit), check.attributes = FALSE)
```

```
[1] TRUE
```

d) Variation explained

```r
c(R2 = summary(fit)$r.squared, adjR2 = summary(fit)$adj.r.squared)
```

```
      R2    adjR2
0.4546766 0.4462870
```

The covariates explain about **45.47%** (unadjusted R^2) of the variation in salary. The adjusted R^2(**44.63%**) penalizes the number of input features. It is the better choice for comparing models of different sizes or for estimating the difference.

e) Largest absolute residual

```r
i_max <- which.max(abs(resid(fit)))
i_max; resid(fit)[i_max]
```

```
44
44
```

```
     44
99592.03
```

```
Salaries[i_max, ]
```

```
   rank discipline yrs.since.phd yrs.service  sex salary
44 Prof          B            38          38 Male 231545
```

Observation *row 44* has the largest absolute residual (about + *+$99,592*).

f) 99% CIs

```
confint(fit, c("yrs.since.phd", "yrs.service"), level = 0.99)
```

```
                   0.5 %     99.5 %
yrs.since.phd   -88.75365 1158.87021
yrs.service   -1038.11487   59.08344
```

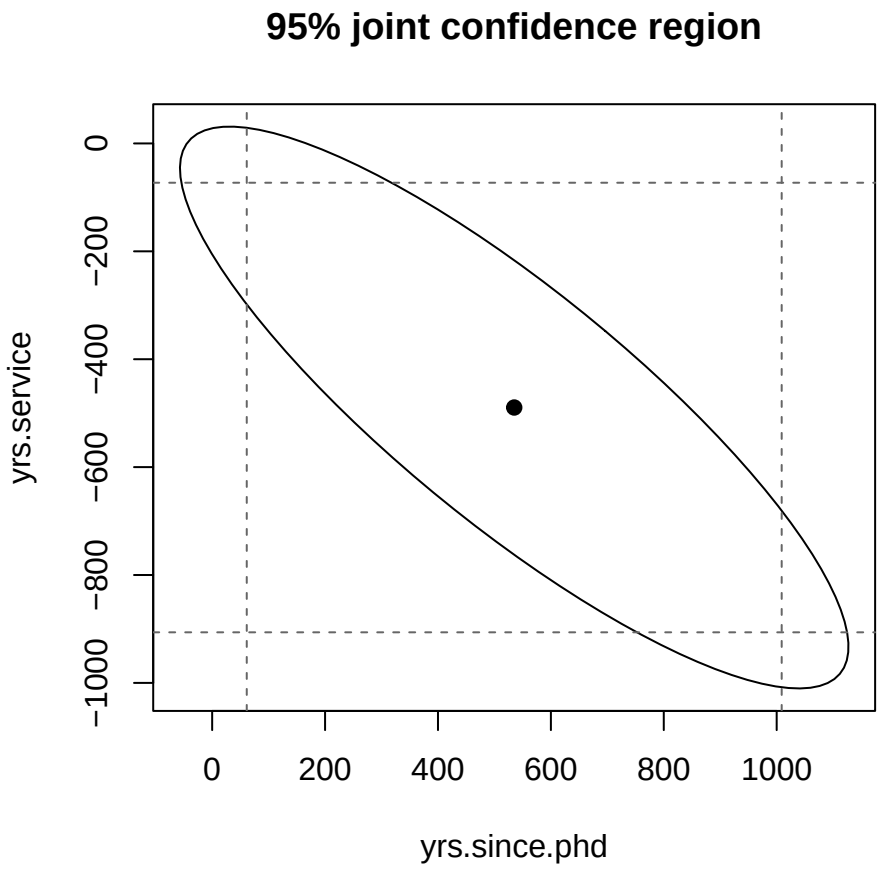
```
confint(fit, c("yrs.since.phd", "yrs.service"), level = 0.95)
```

```
                  2.5 %     97.5 %
yrs.since.phd   61.24805 1008.86852
yrs.service   -906.19882  -72.83261
```

Both 99% confidence intervals of years since phd and years service contain 0. Thus even these coefficients are significant at 5% but not at 1%.

g) 95% joint confidence region

```
idx <- c("yrs.since.phd", "yrs.service")
cr <- ellipse(fit, which = idx, level = 0.95)
ci <- confint(fit, idx, level = 0.95)
plot(cr, type = "l", xlab = idx[1], ylab = idx[2],
     main = "95% joint confidence region")
points(coef(fit)[idx[1]], coef(fit)[idx[2]], pch = 19)
abline(v = ci[1, ], h = ci[2, ], lty = 2, col = "grey40")
```



The 95% joint confidence region for years of service and years since PhD is an ellipse. Since these two variables are positively correlated, their coefficient estimates tend to be negatively correlated, making the ellipse elongated and directed downward to the right(this ellipse only can reflect the correlation of esitimate coefficient, not correlation between variables). This means we can estimate their combined effect more precisely, but it is difficult to separate their individual effects.

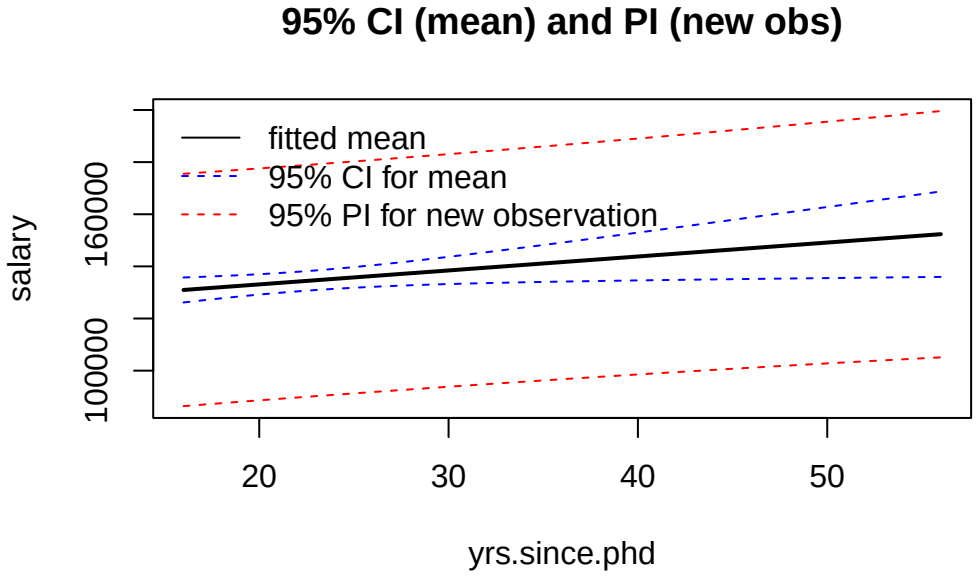
h) Pointwise and simultaneous 95% bands

I take yrs.since.phd as the main variable of interest, letting it range from the median of yrs.service (16 years) to its maximum, and hold the other covariates fixed at rank = Prof, discipline = B, sex = Male, and yrs.service at its median.

```
new <- data.frame(yrs.since.phd = median(Salaries$yrs.service):max(Salaries$yrs.since.phd),
                  yrs.service   = median(Salaries$yrs.service),
                  rank = "Prof", discipline = "B", sex = "Male")
```

```r
ci <- predict(fit, new, interval = "confidence", level = 0.95)
pi <- predict(fit, new, interval = "prediction", level = 0.95)

x <- new$yrs.since.phd
plot(x, ci[, "fit"], type = "l", lwd = 2, ylim = range(pi),
     xlab = "yrs.since.phd", ylab = "salary",
     main = "95% CI (mean) and PI (new obs)")
lines(x, ci[, "lwr"], col = "blue", lty = 2); lines(x, ci[, "upr"], col = "blue", lty = 2)
lines(x, pi[, "lwr"], col = "red",  lty = 2); lines(x, pi[, "upr"], col = "red",  lty = 2)
legend("topleft", bty = "n", lty = c(1, 2, 2), col = c("black", "blue", "red"),
       legend = c("fitted mean", "95% CI for mean", "95% PI for new observation"))
```



Prediction bands are much wider than mean bands because they include the individual random error σ^2:

$$\text{CI: } \hat{y}_0 \pm t_{0.975,\,n-p}\,\hat{\sigma}\sqrt{x_0^T(X^TX)^{-1}x_0}, \qquad \text{PI: } \hat{y}_0 \pm t_{0.975,\,n-p}\,\hat{\sigma}\sqrt{1 + x_0^T(X^TX)^{-1}x_0}$$

i) Partial R^2 for `yrs.since.phd`

```r
fit_red <- update(fit, . ~ . - yrs.since.phd)
partialR2 <- (deviance(fit_red) - deviance(fit)) / deviance(fit_red)
partialR2
```

```
[1] 0.01248159
```

Partial R^2(~0.0125) for yrs.since.phd presents if we add variable yrs.since.phd, 1.25% of rest variance can be explianed. in other words, that means the independent function of feature yrs.since.phd.

j) EHW (heteroskedasticity-consistent) standard errors

```r
se_lm  <- sqrt(diag(vcov(fit)))
se_ehw <- sqrt(diag(vcovHC(fit, type = "HC0")))
cbind(lm = se_lm, EHW = se_ehw)
```

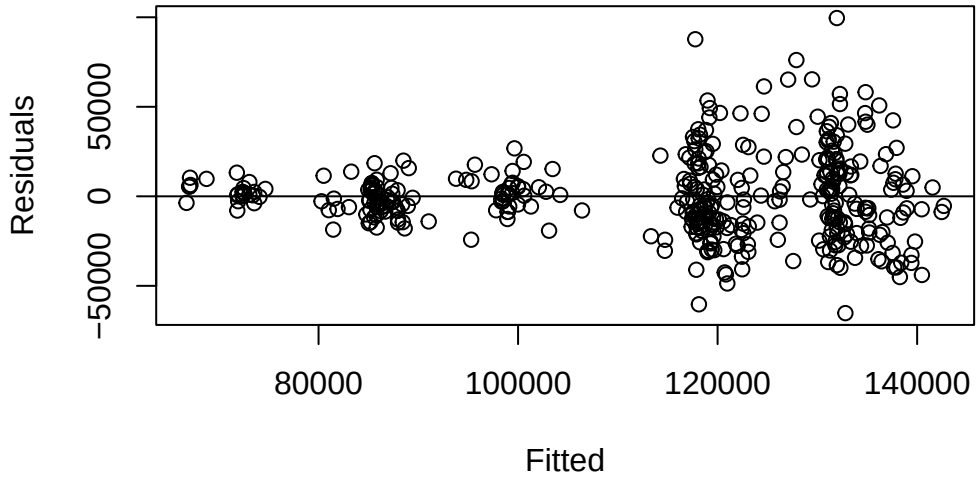
```
                    lm       EHW
(Intercept)   4588.6009 2870.1932
rankAssocProf 4145.2783 2186.1421
rankProf      4237.5233 3255.9927
disciplineB   2342.8753 2295.7952
yrs.since.phd  240.9941  309.6959
yrs.service    211.9376  304.1022
sexMale       3858.6684 2374.7035
```

```r
coeftest(fit, vcov. = vcovHC(fit, type = "HC0"))
```

```
t test of coefficients:

              Estimate Std. Error t value  Pr(>|t|)
(Intercept)   65955.23    2870.19 22.9794 < 2.2e-16 ***
rankAssocProf 12907.59    2186.14  5.9043  7.71e-09 ***
rankProf      45066.00    3255.99 13.8409 < 2.2e-16 ***
disciplineB   14417.63    2295.80  6.2800  9.04e-10 ***
yrs.since.phd   535.06     309.70  1.7277   0.08484 .
yrs.service    -489.52     304.10 -1.6097   0.10827
sexMale        4783.49    2374.70  2.0144   0.04466 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
plot(fitted(fit), resid(fit), xlab = "Fitted", ylab = "Residuals"); abline(h = 0)
```




```
bptest(fit)
```

```
    studentized Breusch-Pagan test

data:  fit
BP = 65.055, df = 6, p-value = 4.205e-12
```

From the residuals-vs-fitted plot (the spread of residuals widens as fitted salary increases) and the Breusch–Pagan test (p 4e-12), we conclude there is significant heteroskedasticity. We therefore use the EHW (HC0) estimator to obtain heteroskedasticity-robust standard error.

Problem 2 (25 pt)

A series of $n+1$ observations y_i $(i = 1, \dots, n+1)$ are taken from a normal distribution with unknown variance σ^2. After the first n observations it is suspected that there is a sudden change in the mean of the distribution. That is, assume the first n observations are iid $y_1, ..., y_n \sim N(\mu, \sigma^2)$ the $y_{(n+1)} \sim N(\mu + \delta, \sigma^2)$.

a) Matrix form

$$\underbrace{\begin{pmatrix} y_1 \\ \vdots \\ y_n \\ y_{n+1} \end{pmatrix}}_{y} = \underbrace{\begin{pmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 1 & 1 \end{pmatrix}}_{X} \underbrace{\begin{pmatrix} \mu \\ \delta \end{pmatrix}}_{\beta} + \epsilon, \qquad \epsilon \sim N(0, \sigma^2 I_{n+1}).$$

b) LS estimates

$$X^T X = \begin{pmatrix} n+1 & 1 \\ 1 & 1 \end{pmatrix}, \quad (X^T X)^{-1} = \frac{1}{n} \begin{pmatrix} 1 & -1 \\ -1 & n+1 \end{pmatrix}, \quad X^T y = \begin{pmatrix} \sum_{i=1}^{n+1} y_i \\ y_{n+1} \end{pmatrix}.$$

So $\hat{\beta} = (X^T X)^{-1} X^T y$ gives

$$\hat{\mu} = \frac{1}{n}\Big(\sum_{i=1}^{n+1} y_i - y_{n+1} \Big) = \bar{y}_n = \frac{1}{n} \sum_{i=1}^{n} y_i, \qquad \hat{\delta} = \frac{1}{n}\Big(-\sum_{i=1}^{n+1} y_i + (n+1) y_{n+1} \Big) = y_{n+1} - \bar{y}_n.$$

c) Test statistic for $H_0 : \delta = 0$

We want to test whether the $(n+1)$st observation has the same mean as the first n:

$$H_0 : \delta = 0 \qquad \text{vs.} \qquad H_1 : \delta \neq 0.$$

The natural test statistic compares the estimated jump $\hat{\delta}$ to its standard error.

Step 1: From (b), $\hat{\delta} = y_{n+1} - \bar{y}_n$.

Step 2: Since y_{n+1} and $\bar{y}_n$ are independent,

$$\mathrm{Var}(\hat{\delta}) = \mathrm{Var}(y_{n+1}) + \mathrm{Var}(\bar{y}_n) = \sigma^2 + \frac{\sigma^2}{n} = \sigma^2 \left(1 + \frac{1}{n} \right).$$

Step 3: Because σ^2 is unknown, we estimate it from the first n observations, which all share the same mean μ:

$$\hat{\sigma}^2 = s_n^2 = \frac{1}{n-1} \sum_{i=1}^{n} (y_i - \bar{y}_n)^2.$$

Step 4: the test statistic. Dividing the estimate by its estimated standard error gives

$$T = \frac{\hat{\delta}}{\hat{\sigma}\sqrt{1+1/n}} = \frac{y_{n+1} - \bar{y}_n}{s_n\sqrt{1+1/n}} \overset{H_0}{\sim} t_{n-1}$$

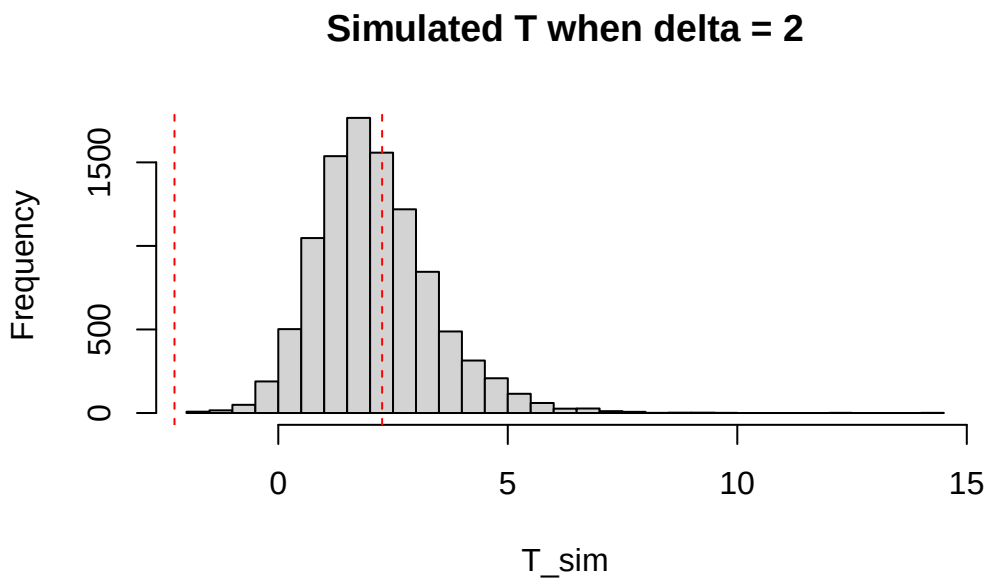
We reject H_0 at level α (two-sided) if $|T| > t_{n-1,\,1-\alpha/2}$. Intuitively, T measures how many standard errors the new observation lies from the previous average; if this is too large to be explained by normal random variation, we conclude that the mean has shifted.

d) Power by simulation ($\sigma^2 = 1$, $\delta = 2$)

```r
n <- 10; delta <- 2; B <- 10000
crit <- qt(0.975, df = n - 1)

T_sim <- replicate(B, {
  y <- c(rnorm(n, 0, 1), rnorm(1, delta, 1))
  (y[n + 1] - mean(y[1:n])) / (sd(y[1:n]) * sqrt(1 + 1/n))
})

hist(T_sim, breaks = 50, main = "Simulated T when delta = 2")
abline(v = c(-crit, crit), col = "red", lty = 2)
```




```r
mean(abs(T_sim) > crit)
```

```
[1] 0.4038
```

We set $n = 10$ and $\mu = 0$ (the value of μ does not affect T), and simulate $B = 10{,}000$ datasets with $\sigma^2 = 1$ and $\delta = 2$. For a two-sided test at the 5% level, we reject H_0 when $|T| > t_{9,\,0.975} \approx 2.26$. The simulated power is about **0.4003**, meaning that even when the mean truly shifts by 2 standard deviations, the test detects it only about 40% of the time.

Problem 3 (25 pt)

AI use disclosure: For this problem, I used an AI assistant to help write the simulation code. I went through each step and understand what it does and why.

Simulation setup

Step 1. Choose the true model. I use $n = 20$ observations and two covariates. The true model is

$$y = 1 + 0 \cdot x_1 + 0.5 \cdot x_2 + \varepsilon,$$

So $\beta = (\beta_0, \beta_1, \beta_2) = (1, 0, 0.5)$. I need both zero and nonzero coefficients because part (c) asks for two different things. If every coefficient were nonzero, every rejection would be correct, and I could not measure the Type I error. If every coefficient were zero, every rejection would be wrong, and I could not measure power. So I set $\beta_1 = 0$: here H_0 is true, and its rejection rate is the Type I error. For $\beta_0 = 1$ and $\beta_2 = 0.5$, H_0 is false, and their rejection rates are the power.

Step 2. Generate data. In each simulation I draw new errors ε, compute y, and fit the regression with `lm()`.

Step 3. Record three things. For each coefficient I save:

- the statistic $(\hat{\beta}_j - \beta_j)/SE(\hat{\beta}_j)$, for part (a);
- whether the 95% confidence interval contains the true β_j, for part (b);
- whether the p-value for $H_0 : \beta_j = 0$ is below 0.05, for part (c).

Step 4. Repeat. I repeat Steps 2–3 $B = 5000$ times and summarize the results.

Step 5. Change the errors. I run the same simulation three times with different error distributions: normal, heavy-tailed, and skewed. All three have mean 0 and variance 1, so they differ only in shape.

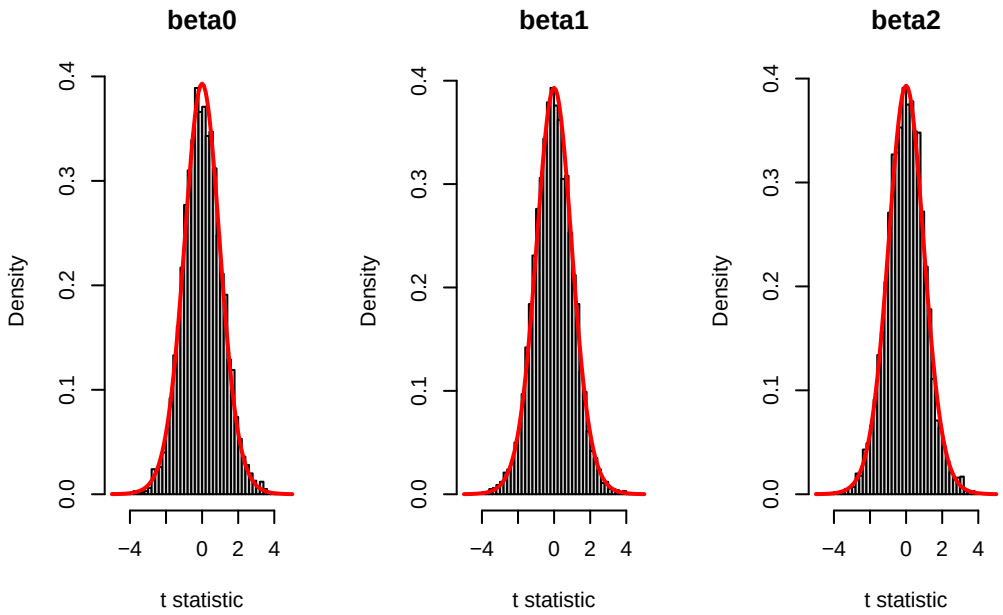
```r
set.seed(220)
n <- 20                    # sample size
B <- 5000                  # number of replications
beta <- c(1, 0, 0.5)       # true coefficients
x1 <- runif(n)             # covariates, fixed across replications
x2 <- rnorm(n)

# make_error: a function that generates n random errors
run_sim <- function(make_error) {
  tstat  <- matrix(NA, B, 3)   # (a) t statistics
  cover  <- matrix(NA, B, 3)   # (b) does the CI contain the true beta?
  reject <- matrix(NA, B, 3)   # (c) do we reject H0: beta = 0?
  for (b in 1:B) {
    y   <- beta[1] + beta[2] * x1 + beta[3] * x2 + make_error(n)
    fit <- lm(y ~ x1 + x2)
    est <- coef(fit)
    se  <- sqrt(diag(vcov(fit)))
    ci  <- confint(fit, level = 0.95)
    tstat[b, ]  <- (est - beta) / se
    cover[b, ]  <- (ci[, 1] <= beta) & (beta <= ci[, 2])
    reject[b, ] <- summary(fit)$coefficients[, 4] < 0.05
  }
  list(tstat = tstat, coverage = colMeans(cover), reject = colMeans(reject))
}
```

a) Distribution of the t statistic (normal errors)

```r
normal <- run_sim(function(n) rnorm(n))

par(mfrow = c(1, 3))
for (j in 1:3) {
  hist(normal$tstat[, j], breaks = 50, freq = FALSE, xlim = c(-5, 5),
       main = paste0("beta", j - 1), xlab = "t statistic")
  curve(dt(x, df = n - 3), add = TRUE, col = "red", lwd = 2)
}
```




```
par(mfrow = c(1, 1))
```

The histograms show the 5000 t statistics I got from the simulation. The red curve is the t distribution with $n - p = 20 - 3 = 17$ degrees of freedom. For all three coefficients, the histogram and the red curve have the same shape, so the simulated statistics really do follow t_{17}. This matches the theory: when the errors are normal, the standardized statistic $(\hat{\beta}_j - \beta_j)/SE(\hat{\beta}_j)$ follows a t_{n-p} distribution.

b) Coverage of the confidence intervals

```
normal$coverage
```

```
[1] 0.9484 0.9506 0.9484
```

The coverage is about 0.95 for every coefficient (0.948, 0.951, 0.948). With 5000 replications, the Monte Carlo error is about ± 0.006, so these values are consistent with the nominal 95%. The intervals have the desired coverage.

c) Type I error and power

```
normal$reject
```

```
[1] 0.7118 0.0494 0.6750
```

- **Type I error:** For $\beta_1 = 0$, the null is true, and we reject it in about 4.9% of simulations.
- **Power:** For $\beta_0 = 1$ and $\beta_2 = 0.5$, the null is false. We reject it about 71% and 68% of the simulations.

d) Non-normal errors

AI use disclosure: I used an AI to explain how the heavy-tailed and skewed error distributions check works, and how to read the results. I then checked the output and rewrote the explanation in my own words.

I repeat the simulation with two other error distributions, specifically,

```
heavy  <- run_sim(function(n) rt(n, df = 3) / sqrt(3)) # ai help me design that
skewed <- run_sim(function(n) rexp(n) - 1) # ai help me design that

results <- rbind(normal = c(normal$coverage, normal$reject),
                 heavy  = c(heavy$coverage,  heavy$reject),
                 skewed = c(skewed$coverage, skewed$reject))
colnames(results) <- c("cover_b0", "cover_b1", "cover_b2",
                       "reject_b0", "reject_b1", "reject_b2")
round(results, 3)
```

```
       cover_b0 cover_b1 cover_b2 reject_b0 reject_b1 reject_b2
normal    0.948    0.951    0.948     0.712     0.049     0.675
heavy     0.958    0.959    0.952     0.779     0.041     0.761
skewed    0.943    0.952    0.946     0.765     0.048     0.713
```

```
# How often the t statistic for beta0 falls in each tail
# (each should be about 2.5% if the t distribution is correct)
q <- qt(0.975, df = n - 3)
tails <- rbind(normal = c(mean(normal$tstat[, 1] < -q), mean(normal$tstat[, 1] > q)),
               heavy  = c(mean(heavy$tstat[, 1]  < -q), mean(heavy$tstat[, 1]  > q)),
               skewed = c(mean(skewed$tstat[, 1] < -q), mean(skewed$tstat[, 1] > q)))
colnames(tails) <- c("lower tail", "upper tail")
round(tails, 3)
```

```
        lower tail upper tail
normal       0.022      0.029
heavy        0.022      0.020
skewed       0.046      0.011
```

Heavy-tailed errors. The results are still very similar to normal errors. The coverage is close to 95%, and the Type I error is close to 5%. The rejection rates in the left and right tails are also similar (2.2% vs. 2.0%), suggesting that the t-test still works well.

Skewed errors. At first the results also look fine: coverage is about 95% (0.943, 0.952, 0.946) and the Type I error for β_1 is 4.8%. However, the rejection rates in the two tails are very different (4.6% vs. 1.1%), suggesting that the t statistic is no longer symmetric.

Which is worse? Skewed errors are worse. With heavy-tailed errors, the rejection rates in the left and right tails are similar (2.2% vs. 2.0%), suggesting that the shape of the t-statistic distribution has not changed much and the t-test still works well. However, with skewed errors, the rejection rates in the two tails are very unbalanced (4.6% vs. 1.1%), indicating that the distribution of the t statistic has become skewed and the (t_{17}) distribution can no longer accurately describe it.

Problem 4 (15 pts)

a) Divorce rate over time

The official introduction of `divusa` data

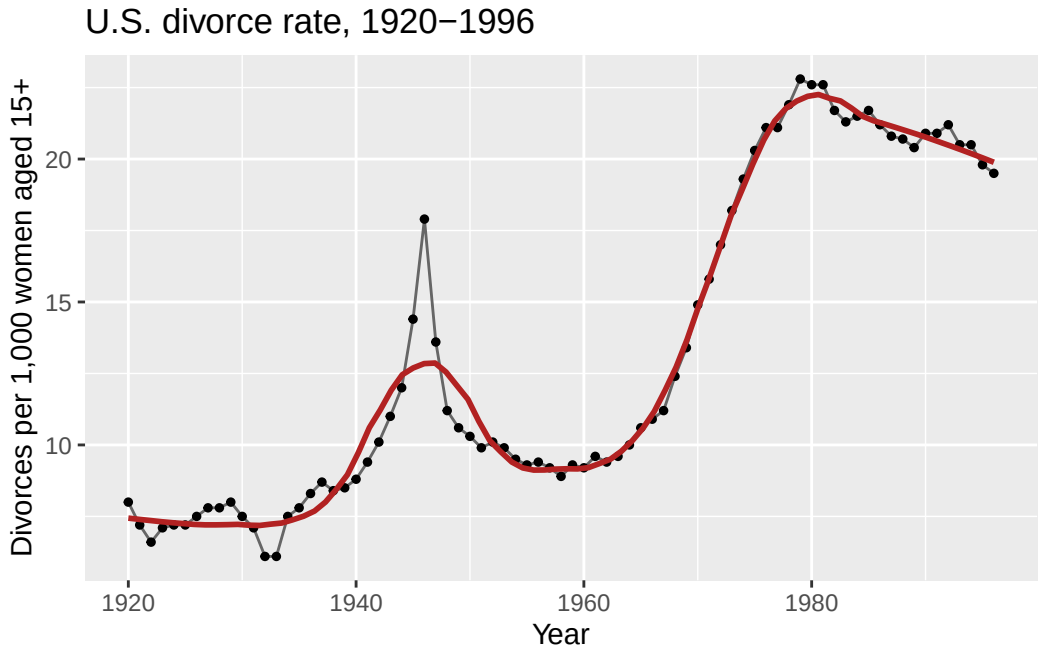
Variable	Description
`year`	the year from 1920–1996
`divorce`	divorces per 1,000 women aged 15 or more
`unemployed`	unemployment rate
`femlab`	percent female participation in labor force aged 16+
`marriage`	marriages per 1,000 unmarried women aged 16+
`birth`	births per 1,000 women aged 15–44
`military`	military personnel per 1,000 population

```
data(divusa)
head(divusa)
```

```
  year divorce unemployed femlab marriage birth military
1 1920     8.0        5.2  22.70     92.0 117.9   3.2247
2 1921     7.2       11.7  22.79     83.0 119.8   3.5614
3 1922     6.6        6.7  22.88     79.7 111.2   2.4553
4 1923     7.1        2.4  22.97     85.2 110.5   2.2065
```

```
5 1924     7.2         5.0  23.06     80.3 110.9   2.2889
6 1925     7.2         3.2  23.15     79.2 106.6   2.1735
```

```
ggplot(divusa, aes(year, divorce)) +
  geom_line(colour = "grey40") + geom_point(size = 1) + geom_smooth(method = "loess", se = F
  labs(x = "Year", y = "Divorces per 1,000 women aged 15+",
       title = "U.S. divorce rate, 1920-1996")
```



The divorce rate is **not steady**. It rises overall, from about 8 per 1,000 in 1920 to a peak of about 23 around 1980, then declines slightly.

b) Regression and the `year` coefficient

```
fit4 <- lm(divorce ~ year + unemployed + femlab + marriage + birth + military, data = divusa
summary(fit4)
```

```
Call:
lm(formula = divorce ~ year + unemployed + femlab + marriage +
    birth + military, data = divusa)

Residuals:
    Min      1Q  Median      3Q     Max
```

```
-2.9087 -0.9212 -0.0935  0.7447  3.4689

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 380.14761   99.20371   3.832 0.000274 ***
year         -0.20312    0.05333  -3.809 0.000297 ***
unemployed   -0.04933    0.05378  -0.917 0.362171
femlab        0.80793    0.11487   7.033 1.09e-09 ***
marriage      0.14977    0.02382   6.287 2.42e-08 ***
birth        -0.11695    0.01470  -7.957 2.19e-11 ***
military     -0.04276    0.01372  -3.117 0.002652 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.513 on 70 degrees of freedom
Multiple R-squared:  0.9344,    Adjusted R-squared:  0.9288
F-statistic: 166.2 on 6 and 70 DF,  p-value: < 2.2e-16
```

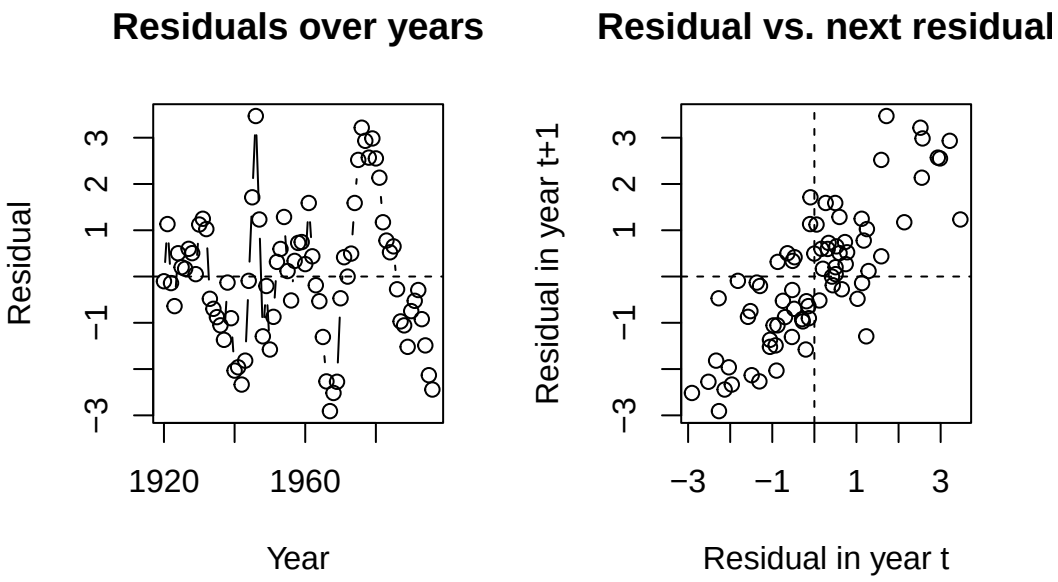
The coefficient on `year` is **-0.20312**. Holding unemployment, female labor-force participation, marriage rate, birth rate and military size fixed, each additional year is associated with about 0.20 *fewer* divorces per 1,000 women aged 15+.

This seems inconsistent with (a). My guess is that the difference may come from whether other variables are controlled for. The plot shows the overall trend in divorce rates over time, while the regression examines the relationship between year and divorce rate after controlling for other factors. Therefore, the two may show opposite results.

c) Correlated errors

This is time series data, so the errors are likely autocorrelated. Some factors that affect divorce are not in the model, such as income level or someo other social features. And their effects in previous year may transfer to the next as well. Since they are left out of the model, their influence ends up in the errors.

```r
e <- resid(fit4)
n <- length(e)
par(mfrow = c(1, 2))
plot(divusa$year, e, type = "b", xlab = "Year", ylab = "Residual",
     main = "Residuals over years")
abline(h = 0, lty = 2)
plot(e[-n], e[-1], xlab = "Residual in year t", ylab = "Residual in year t+1",
     main = "Residual vs. next residual")
abline(h = 0, v = 0, lty = 2)
```




```
par(mfrow = c(1, 1))
```

Conclusion: the errors are strongly positively autocorrelated and the OLS error independence assumption fails, the ols result are not very reliable.

d) Formal test

```
dwtest(fit4)
```

```
    Durbin-Watson test

data:  fit4
DW = 0.37429, p-value < 2.2e-16
alternative hypothesis: true autocorrelation is greater than 0
```

The Durbin–Watson statistic is 0.37429, with a p-value of 2.2e-16. Thus, we reject the null hypothesis (H_0) of no autocorrelation and conclude that there is strong error autocorrelation in our model.